

BCSE497J - Project-I

WEARABLE PERIORBITAL MULTI-SENSOR SYSTEM FOR EARLY DETECTION OF DIABETIC NEURO-RETINAL CHANGES

23BDS0316 SHLOK GOENKA

23BCI0020 ARYAN HUNDIA

Under the Supervision of

Dr. JAYASHREE J

Associate Professor Grade 2

School of Computer Science and Engineering (SCOPE)

B.Tech.

in

Computer Science and Engineering

(Specialization in Data Science)

School of Computer Science and Engineering



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

September 2026

ABSTRACT

Diabetes mellitus affects approximately 77 million adults in India, a figure projected to reach 125 million by 2045. Individuals with diabetes of ten or more years duration develop two microvascular complications in parallel: diabetic retinopathy (DR), affecting an estimated 12.5 percent of the diabetic population nationally, and diabetic peripheral neuropathy (DPN), reported in 18.8 to 61.9 percent of Indian patients. Both complications originate from the same pathophysiological process, namely chronic hyperglycaemia damaging small blood vessels and nerve fibres. Despite this shared aetiology, the two conditions are screened independently, requiring separate specialists, separate instrumentation and separate clinical visits. Screening is periodic rather than continuous, so deterioration occurring between annual examinations remains undetected.

This project proposes a wearable periorbital multi-sensor system implemented on a spectacle frame that monitors physiological markers of both retinopathy and neuropathy simultaneously and continuously. Four complementary sensing modalities are integrated: infrared pupillography for autonomic nerve assessment, contactless periorbital thermography for retinal microvascular perfusion, bioimpedance spectroscopy for periorbital fluid accumulation associated with macular oedema, and photoplethysmography at the superficial temporal artery for vascular stiffness and cardiac autonomic function. The periorbital region is selected because it is the sole external anatomical site at which the relevant autonomic nerves, retinal blood supply, venous drainage and systemic vasculature converge within a few centimetres.

An embedded edge-artificial-intelligence processor performs multi-modal sensor fusion, incorporating glycaemic context obtained from a continuous glucose monitor over a wireless link, and computes a composite Neuro-Retinal Deterioration Index. Cross-validation between the four independent signal sources suppresses false positives arising from environmental confounders, a limitation inherent to single-modality devices. The projected component cost of under eight thousand rupees permits deployment at rural primary health centres, where fundus photography equipment is unavailable. The expected outcome is detection of subclinical deterioration substantially earlier than conventional annual screening, together with a unified risk score interpretable by frontline health workers without specialist training.

Keywords - Diabetic Retinopathy, Diabetic Neuropathy, Wearable Biosensors, Periorbital Sensing, Multi-Modal Sensor Fusion, Edge Computing, Point-of-Care Screening.

TABLE OF CONTENTS

Sl.No	Contents	Page No.
	Abstract	i
1.	INTRODUCTION	1
	1.1 Background	1
	1.2 Motivation	2
	1.3 Scope of the Project	3
2.	PROJECT DESCRIPTION AND GOALS	4
	2.1 Literature Review	4
	2.2 Research Gap	7
	2.3 Objectives	8
	2.4 Problem Statement	9
	2.5 Project Plan	10
3.	TECHNICAL SPECIFICATION	11
	3.1 Requirements	11
	3.1.1 Functional	11
	3.1.2 Non-Functional	12
	3.2 Feasibility Study	13
	3.2.1 Technical Feasibility	13
	3.2.2 Economic Feasibility	14
	3.2.3 Social Feasibility	15
	3.3 System Specification	16
	3.3.1 Hardware Specification	16
	3.3.2 Software Specification	17
4.	DESIGN APPROACH AND DETAILS	18
	4.1 System Architecture	18
	4.2 Design	20

4.2.1 Data Flow Diagram	20
4.2.2 Use Case Diagram	21
5. REFERENCES	23

1. INTRODUCTION

1.1 Background

Diabetes mellitus has become one of the most significant public health challenges in India. Current estimates place the diabetic population at approximately 77 million adults, with projections indicating a rise to 125 million by 2045. Sustained hyperglycaemia progressively damages the smallest blood vessels and the peripheral nervous system, producing a cluster of microvascular complications that develop silently over many years.

Two of these complications dominate the clinical burden. Diabetic retinopathy, with a nationally weighted prevalence of 12.5 percent and vision-threatening disease in 4.0 percent, is a leading cause of preventable blindness. Diabetic peripheral neuropathy, reported across Indian studies at prevalences ranging from 18.8 to 61.9 percent, causes sensory loss, foot ulceration and, in advanced cases, amputation. Both arise from the same underlying pathology, yet clinical practice treats them as unrelated conditions.

Screening for retinopathy requires a fundus camera costing between two and fifteen lakh rupees, operated by trained personnel and interpreted by an ophthalmologist. Neuropathy assessment relies on monofilament testing, biothesiometry or nerve conduction studies performed by a different specialist. Neither pathway communicates with the other, and both are conducted at annual intervals at best. In rural India, where the density of ophthalmologists falls below four per hundred thousand population, this infrastructure is largely absent, and a substantial proportion of patients never undergo screening at all.

1.2 Motivation

The motivation for this project arises from a specific and clinically documented gap. Patients with long-standing diabetes develop retinopathy and neuropathy concurrently, and Indian clinical studies have established a statistically significant association between the severity of the two conditions. Despite this correlation, no screening instrument assesses both simultaneously.

A second consideration is the timing of detection. Conventional fundus photography identifies structural lesions such as microaneurysms and haemorrhages, which appear only after irreversible retinal tissue damage has occurred. Published evidence demonstrates that functional physiological changes precede these structural findings. Ocular surface temperature falls measurably in non-proliferative retinopathy relative to matched controls, reflecting reduced microvascular perfusion. Pupillary light reflex parameters degrade in a large majority of diabetic patients, including individuals in whom standard neuropathy tests return normal results. These observations indicate a substantial diagnostic window that current practice does not exploit.

The third and most practical motivation concerns access. A screening device that costs a small fraction of a fundus camera, requires no specialist operator and produces an interpretable risk

classification could be deployed at primary health centres and operated by accredited social health activists. This shifts diabetic complication screening from a scarce, urban, specialist-dependent service toward a distributed and continuous one.

1.3 Scope of the Project

This project encompasses the design, development and validation of a wearable periorbital multi-sensor platform for the combined monitoring of diabetic retinopathy and neuropathy indicators. The work includes selection and justification of four sensing modalities on the basis of published clinical evidence, design of the sensor placement geometry on a spectacle-frame carrier, and specification of the signal acquisition and conditioning electronics.

The computational scope covers the development of a multi-modal fusion algorithm that combines the four sensor streams with glycaemic context to produce a composite Neuro-Retinal Deterioration Index, together with the associated feature extraction, preprocessing and threshold-based classification logic. Model development and statistical validation are undertaken using cohorts constructed from published clinical parameter distributions prior to hardware availability.

The scope excludes clinical diagnosis. The device produces a risk index intended to prompt referral, not a diagnostic determination, which remains the responsibility of a qualified clinician. Regulatory certification, large-scale multi-centre clinical trials and commercial manufacture lie outside the present project boundary, as does treatment or therapeutic intervention of any kind.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

The literature relevant to this project spans four sensing domains together with the clinical evidence linking diabetic retinopathy and neuropathy. Table 1 summarises the principal studies examined, their methodologies, findings and limitations.

Ref.	Author / Year	Methodology	Key Findings	Limitations
[1]	Chandrasekar et al. (2021)	Infrared ocular surface thermography using desktop thermal camera; 5 measurement points per eye	NPDR eyes 0.61 ± 0.12 °C cooler at cornea; PDR 0.62 ± 0.11 °C cooler; $p < 0.001$	Desktop instrument; single modality; no continuous monitoring
[2]	Sodi et al. (2009)	Ocular surface temperature measurement across corneal and conjunctival sites	Significantly lower temperature at all five points in NPDR versus controls	Clinic-based; no neuropathy assessment
[3]	Dütsch et al. (2004)	Desktop pupillography assessing autonomic function in type 2 diabetes	77.8 % of patients showed abnormal pupillography versus 56 % detected by standard tests	Not wearable; retinopathy not evaluated
[4]	Ferrari et al. (2020)	Pupillary light reflex dynamics in diabetic retinal neuropathy	Constriction and dilation velocities significantly reduced before observable retinopathy signs	Laboratory setting; no fusion with other modalities
[5]	Hwang et al. (2023)	Whole-body bioimpedance analysis correlated with optical coherence tomography	Phase angle $5.45 \pm 0.84^\circ$ in DME versus $6.69 \pm 0.69^\circ$ in no-DR group; $p < 0.001$	Full-body measurement; confounded by systemic fluid status
[6]	Yao et al. (2022)	Fluid overload assessment via bioimpedance in diabetic macular oedema	ECW/TBW ratio improved discrimination; AUROC increased to 0.826	Not periorbital; requires clinical instrument
[7]	ACCORD Study Group (2022)	Longitudinal HRV analysis in 5,810 participants with type 2 diabetes	SDNN below 8.2 ms associated with hazard ratio 1.32 for vision-threatening retinopathy	Analytical study; no device implementation
[8]	Ko et al. (2014)	Spectacle-mounted pupillometer, 78 g, four-colour LED stimulus	Ten pupil parameters extracted; five differed significantly in diabetic subjects	Single modality; monocular; 30-minute clinic session; never commercialised
[9]	Rasheed et al. (2021)	Cross-sectional clinical correlation study in Indian diabetic cohort	Statistically significant association between retinopathy and neuropathy severity; $p < 0.001$	Observational only; no screening instrument proposed

Ref.	Author / Year	Methodology	Key Findings	Limitations
[10]	Vujosevic et al. (2020)	Review of screening technologies including autonomous artificial intelligence systems	Deep learning on fundus images achieves high sensitivity for referable retinopathy	Requires fundus camera; retinopathy only; episodic acquisition

Table 1. Summary of reviewed literature

Thermographic studies consistently demonstrate that ocular surface temperature is reduced in diabetic retinopathy, with the magnitude of reduction increasing with disease severity. This reflects diminished perfusion in the ophthalmic arterial supply. All reported implementations, however, employ desktop thermal cameras in controlled clinical environments.

Pupillographic research establishes that autonomic denervation of the iris precedes and exceeds the detection capability of conventional neuropathy tests. The work of Dütsch and colleagues is particularly significant, since it identified abnormalities in a substantially larger fraction of patients than standard peripheral and cardiac autonomic testing. Ferrari and colleagues further demonstrated that pupillary velocity parameters degrade before retinopathy becomes clinically observable.

Bioimpedance investigations link fluid distribution to diabetic macular oedema, with phase angle emerging as the most discriminating single parameter. These studies employed whole-body analysers, and the resulting measurements are therefore influenced by cardiac, renal and limb fluid status unrelated to the eye.

Heart rate variability derived from cardiovascular monitoring has been shown in a large longitudinal cohort to predict incident vision-threatening retinopathy, establishing cardiac autonomic dysfunction as a precursor rather than merely a correlate. No wearable device presently exploits this relationship for ocular risk assessment.

The single most relevant precedent is the spectacle-mounted pupillometer of Ko and colleagues, which demonstrated the mechanical feasibility of ocular sensing on an eyewear platform. That device is limited to a single modality, operates monocularly during a supervised clinic session, does not assess retinopathy, incorporates no glycaemic context, and was never patented or commercialised.

2.2 Research Gap

Analysis of the literature, patent databases and government-funded project records identifies the following gaps.

- **Absence of combined assessment.** Every reviewed instrument evaluates either retinopathy or neuropathy. No device assesses both, despite the documented clinical association between them.
- **Dependence on capital-intensive instrumentation.** Retinopathy screening universally requires fundus photography, and neuropathy assessment requires specialist equipment. Neither is available at the primary care level in rural India.

- **Episodic rather than continuous acquisition.** Existing methods capture a single time point at annual intervals, providing no visibility of deterioration occurring between examinations.
- **Absence of a wearable periorbital platform.** Searches of patent and literature databases covering 2020 to 2025 returned no wearable device combining periorbital pupillography, thermography, bioimpedance and photoplethysmography.
- **No unified risk representation.** No published work produces a single composite index integrating retinal, neural and glycaemic indicators into an output interpretable by non-specialist personnel.
- **Untranslated biomarkers.** Periorbital thermography, localised bioimpedance and temporal-artery photoplethysmography have each been validated in research settings but none has been miniaturised into a wearable form factor for diabetic complication monitoring.

2.3 Objectives

The project pursues the following objectives, each defined to be specific, measurable and achievable within the project duration.

- **Objective 1.** To design a spectacle-frame sensing platform integrating four modalities, namely infrared pupillography, periorbital thermography, bioimpedance spectroscopy and photoplethysmography, with a total component cost below eight thousand rupees.
- **Objective 2.** To define the sensor placement geometry and signal acquisition architecture, resolving bus addressing, analog conditioning and camera interfacing constraints within a temple-arm printed circuit board of approximately 12 mm by 60 mm.
- **Objective 3.** To develop a multi-modal fusion algorithm computing a composite Neuro-Retinal Deterioration Index from the four sensor streams together with glycaemic context obtained over a wireless link.
- **Objective 4.** To validate the fusion approach statistically, demonstrating that the composite index achieves greater discrimination between diabetic and control cohorts than any constituent modality evaluated independently, at a significance level of $p < 0.05$.
- **Objective 5.** To implement threshold-based risk stratification producing a three-level classification interpretable by frontline health workers without specialist training.
- **Objective 6.** To document the design comprehensively to support intellectual property protection and subsequent hardware realisation.

2.4 Problem Statement

Patients with diabetes of ten or more years duration develop diabetic retinopathy and diabetic peripheral neuropathy concurrently, driven by a common microvascular and neural pathology.

Current clinical practice screens these complications independently, using distinct instruments operated by distinct specialists at distinct clinical encounters, and does so only at annual intervals. The instrumentation required is capital-intensive and largely unavailable outside urban tertiary centres.

Consequently three failures occur together. Subclinical deterioration detectable through functional physiological markers proceeds unobserved because current methods detect only structural damage. Progression occurring between annual examinations is missed entirely. A large fraction of the diabetic population, particularly in rural areas, receives no screening whatsoever because the necessary infrastructure does not exist.

The problem addressed by this project is therefore the absence of an affordable, continuous, non-invasive instrument capable of monitoring physiological indicators of both diabetic retinopathy and diabetic neuropathy simultaneously, and of expressing the combined result as a single risk measure usable by non-specialist healthcare personnel.

2.5 Project Plan

The project is organised into four sequential phases spanning eleven months. The first phase comprises literature review, prior art analysis and problem formulation. The second addresses sensor selection, dataset construction and development of the fusion model. The third covers circuit design, printed circuit board layout and prototype assembly. The fourth encompasses bench testing, pilot data collection, documentation and intellectual property filing. Figure 1 presents the schedule.

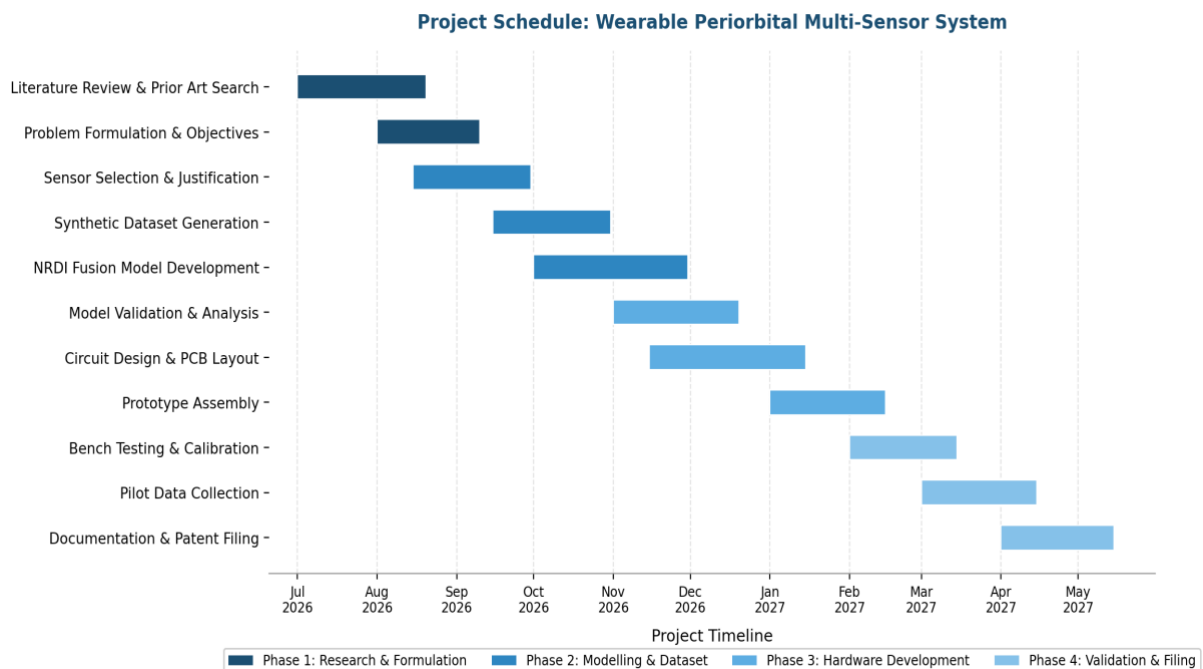


Fig. 1. Gantt chart of project schedule

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional

- **Multi-modal signal acquisition.** The system shall sample four sensing modalities concurrently, acquiring pupillary video at a minimum of 30 frames per second, periorbital temperature at 0.02 °C resolution, tissue impedance across a 1 kHz to 100 kHz frequency sweep, and photoplethysmographic waveforms at a minimum of 100 Hz.
- **Bilateral measurement.** The system shall acquire signals from both left and right periorbital regions independently to permit computation of inter-ocular asymmetry.
- **Signal preprocessing.** The system shall remove baseline drift, power-line interference and motion artefacts from acquired signals prior to feature extraction.
- **Feature extraction.** The system shall derive pupillary dynamic parameters, thermal differentials, impedance phase angle and heart rate variability indices from the conditioned signals.
- **Glycaemic data ingestion.** The system shall establish a wireless connection with a compatible continuous glucose monitor and retrieve time-in-range and glycaemic variability metrics.
- **Composite index computation.** The system shall fuse extracted features with glycaemic context to compute a Neuro-Retinal Deterioration Index on a nought to one hundred scale.
- **Baseline personalisation.** The system shall establish and maintain a per-patient physiological baseline and evaluate subsequent measurements as deviations from that reference.
- **Risk stratification and alerting.** The system shall classify the computed index into three risk categories and activate a haptic notification when a defined threshold is exceeded.
- **Data synchronisation.** The system shall transmit longitudinal records to a remote endpoint for clinician review when network connectivity is available.

3.1.2 Non-Functional

- **Performance.** Index computation shall complete within ninety seconds of an acquisition cycle, executing entirely on the embedded processor without dependence on network availability.
- **Power efficiency.** The system shall sustain a minimum of eight hours of continuous operation from a 250 mAh lithium-polymer cell.

- **Wearability.** Total assembled mass shall remain below one hundred grams, with electronics contained within temple arms of approximately 12 mm by 60 mm.
- **Safety.** All optical emissions shall remain within IEC 62471 photobiological safety limits, and bioimpedance excitation currents shall remain below IEC 60601-1 patient auxiliary current limits.
- **Reliability.** Sensor readings shall remain within specified tolerance across an ambient temperature range of 15 °C to 40 °C, with graceful degradation should any single modality fail.
- **Usability.** Operation shall require no specialist training, with output presented as a three-level colour classification.
- **Data security.** Physiological data shall be encrypted at rest and in transit, with patient identifiers stored separately from measurement records.
- **Maintainability.** Firmware shall be modular by sensor subsystem to permit independent calibration and update.
- **Affordability.** Bill of materials cost shall remain below eight thousand rupees per unit at prototype quantities.

3.2 Feasibility Study

3.2.1 Technical Feasibility

- **Component availability.** All specified sensors are commercially available, well documented and widely used in embedded applications. No custom silicon or unavailable component is required.
- **Evidence base.** Each modality has established clinical validation with published effect sizes. The required measurement resolutions fall well within the specified performance of the selected devices; for example, the reported thermal signal of approximately 0.6 °C exceeds the 0.02 °C resolution of the thermopile by a factor of thirty.
- **Known design challenges.** Three constraints are identified. Fixed I2C addresses on the thermopile and pulse oximetry devices necessitate a bus multiplexer. The parallel camera interface requires the dedicated peripheral available on the selected processor. Component density within the temple-arm footprint requires careful layout, potentially on a flexible substrate.
- **Computational adequacy.** The dual-core 240 MHz processor with integrated wireless connectivity and camera interface provides sufficient capability for the intended preprocessing, feature extraction and inference workload.
- **Team competence.** The required skills in embedded firmware, signal processing and machine learning align with the programme specialisation, supplemented by hardware collaboration for circuit realisation.

3.2.2 Economic Feasibility

- **Development cost.** Prototype component cost is approximately two thousand eight hundred rupees per unit. Printed circuit board fabrication for a small batch adds a further eight hundred rupees. Total development hardware expenditure remains within a modest student project budget.
- **Comparative economics.** A non-mydratic fundus camera costs between two and fifteen lakh rupees and screens only for retinopathy. The proposed device addresses two complications at a small fraction of that capital outlay, altering the economics of primary-care deployment.
- **Return on investment.** Early detection substantially reduces the incidence of vision loss and lower-limb complications. The avoided cost of vitreoretinal surgery, prolonged ulcer management or amputation greatly exceeds the device cost per patient screened.
- **Scalability of production.** All components are standard surface-mount parts amenable to automated assembly, with unit cost decreasing substantially at volume.
- **Funding pathways.** The project aligns with national programmes supporting affordable medical device innovation, and the intended patent filing establishes a basis for translational funding.

3.2.3 Social Feasibility

- **Acceptability to patients.** The device adopts a familiar spectacle form factor and requires no skin penetration, blood sampling or pupil dilation, which supports adherence relative to procedures requiring clinic attendance and mydriasis.
- **Suitability for frontline deployment.** A three-level output classification permits operation by accredited social health activists and auxiliary nurse midwives after brief orientation, addressing the specialist shortage in rural districts.
- **Equity of access.** The cost profile permits distribution to primary health centres presently without any retinopathy screening capability, directly addressing the urban-rural disparity in diabetic care.
- **Ethical considerations.** The device reports risk rather than diagnosis, preserving clinical judgement. Data governance provisions address the confidentiality of continuously acquired physiological information, and informed consent procedures will be established through institutional ethics review.
- **Workforce impact.** The system supplements rather than displaces specialist practice, functioning as a triage instrument that directs scarce ophthalmological capacity toward patients at demonstrated risk.

3.3 System Specification

3.3.1 Hardware Specification

Table 2 details the hardware components, quantities, placements and interfaces.

Component	Part Number	Qty	Placement	Interface	Cost (Rs)
Infrared emitter	OSRAM SFH 4726AS (940 nm)	2	Nose bridge	GPIO / PWM	120
CMOS camera	OV7670	2	Nose bridge	DVP / SCCB	230
Infrared thermopile	MLX90614ESF-BAA	2	Inner orbital rim	I2C	450
Impedance analyser	Analog Devices AD5933	1	Temple-arm PCB	I2C	600
Electrode pads	Ag/AgCl skin contact	4	Upper and lower orbital rim	Analog	100
Analog front-end	AD8606 op-amp buffer	1	Temple-arm PCB	Analog	90
Pulse oximetry sensor	Maxim MAX30102	2	Temple, over temporal artery	I2C	250
I2C multiplexer	TCA9548A	1	Temple-arm PCB	I2C	110
Edge processor	ESP32-S3 (dual-core, 240 MHz)	1	Temple-arm PCB	Host	500
Haptic actuator	LRA C10-100	1	Ear hook	PWM	100
Battery	Li-Po 3.7 V, 250 mAh	1	Temple arm	Power	250
Total	15 components	18	Spectacle-frame platform		2,800

Table 2. Hardware component specification

The development workstation used for model training and firmware compilation comprises an x86-64 processor with a minimum of eight cores, 16 GB of system memory, 512 GB solid-state storage and a CUDA-capable graphics processor for accelerated model training.

3.3.2 Software Specification

- **Operating environment.** FreeRTOS on the embedded target; Ubuntu 22.04 LTS on the development workstation.
- **Programming languages.** C and C++ for firmware; Python 3.11 for signal processing, model development and analysis.
- **Development tools.** ESP-IDF for embedded development, PlatformIO for build management, KiCad for schematic capture and printed circuit board layout, Git for version control.
- **Libraries and frameworks.** NumPy, SciPy and pandas for numerical work; scikit-learn for baseline classifiers; PyTorch for temporal convolutional network development; TensorFlow Lite for Microcontrollers for on-device inference; OpenCV for pupil segmentation; Matplotlib for visualisation.
- **Communication protocols.** Bluetooth Low Energy 5.0 for glucose monitor pairing; Wi-Fi 802.11 b/g/n with MQTT over TLS for cloud synchronisation; I2C for the sensor bus.

- **Data storage.** SQLite for on-device longitudinal records; PostgreSQL for the server-side repository.
- **Security.** AES-256 encryption at rest, TLS 1.3 in transit, and secure boot with signed firmware images.

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

The system is organised into four functional layers. The sensing layer comprises the four modalities distributed across the spectacle frame. The acquisition layer performs bus arbitration, analog conditioning and camera framing. The edge processing layer executes preprocessing, feature extraction, multi-modal fusion and index computation on the embedded processor. The output layer performs risk stratification, local alerting and remote synchronisation. Figure 2 presents this architecture.

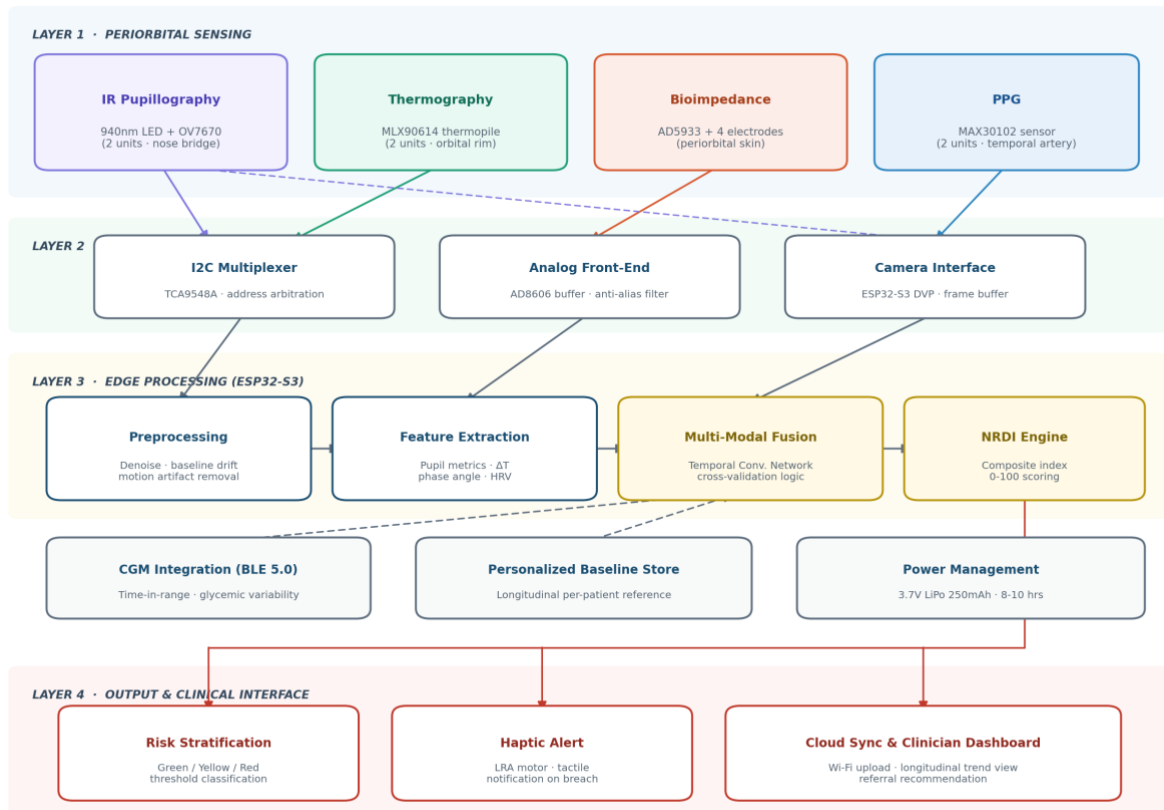


Fig. 2. System architecture

Signals originate at the sensing layer. The two thermopiles and two pulse oximetry sensors share an I2C bus and are addressed through a multiplexer, since each device family presents a fixed address that would otherwise collide. Bioimpedance electrodes connect through an operational amplifier buffer providing impedance matching and anti-alias filtering before digitisation by the impedance analyser. The pupillography cameras connect through the dedicated parallel camera peripheral of the processor.

Within the edge processing layer, preprocessing removes baseline drift and motion artefacts. Feature extraction derives pupillary dynamics, bilateral thermal differentials, impedance phase angle and heart rate variability indices. The fusion stage combines these features with glycaemic context and evaluates them against the stored personalised baseline. The

distinguishing characteristic of this stage is cross-validation between independent modalities: a thermal reduction accompanied by concordant changes in impedance, pupillary response and pulse morphology indicates genuine physiological deterioration, whereas an isolated thermal reduction more probably reflects an environmental confounder. This mechanism suppresses false positives that single-modality devices cannot distinguish.

The output layer converts the computed index into a three-level classification, activates the haptic actuator on threshold breach, and synchronises longitudinal records to a clinician-facing repository when connectivity permits. Figure 3 depicts the physical placement of sensing elements on the spectacle carrier.

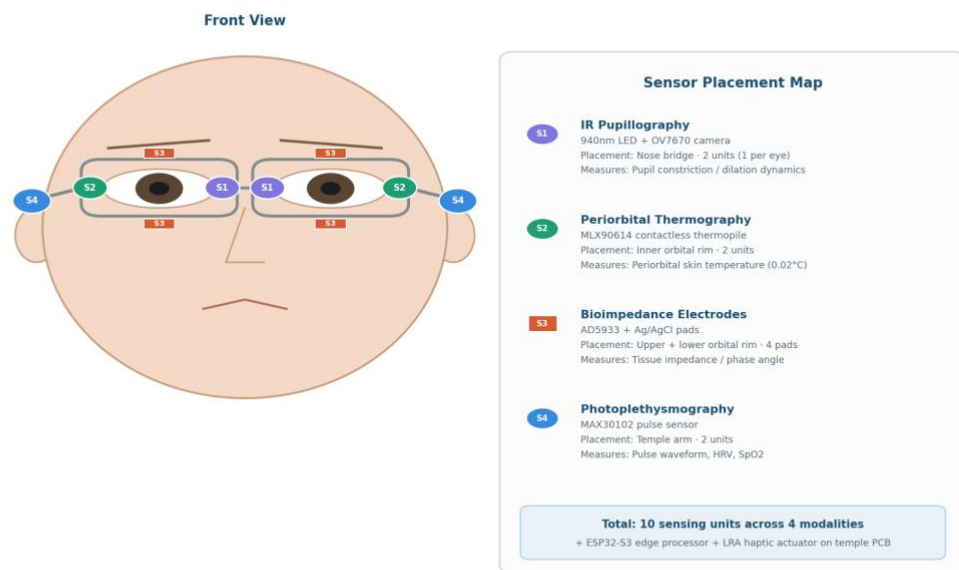


Fig. 3. Sensor placement on the spectacle-frame platform

Placement is determined by underlying anatomy. The nose bridge affords an unobstructed optical path to both pupils at a working distance of two to three centimetres. The inner orbital rim overlies the angular artery, a terminal branch of the ophthalmic artery that also supplies the retina, making it the most informative site for thermal assessment of ocular perfusion. Electrode positions above and below the orbital rim span the periorbital tissue drained by the superior and inferior ophthalmic veins, the pathway through which retinal fluid accumulation manifests externally. The temple placement of the photoplethysmographic sensors overlies the superficial temporal artery, which lies close to the surface and is held in consistent contact by the frame, yielding markedly better signal quality than wrist-based alternatives.

4.2 Design

4.2.1 Data Flow Diagram

Figure 4 presents the level-one data flow diagram. Physiological signals originate from the patient and glycaemic readings from the external monitor. Six processes transform these inputs: signal acquisition, glycaemic ingestion, preprocessing, feature extraction, index computation and output generation. Three data stores retain the raw signal buffer, extracted feature vectors and the personalised patient baseline.

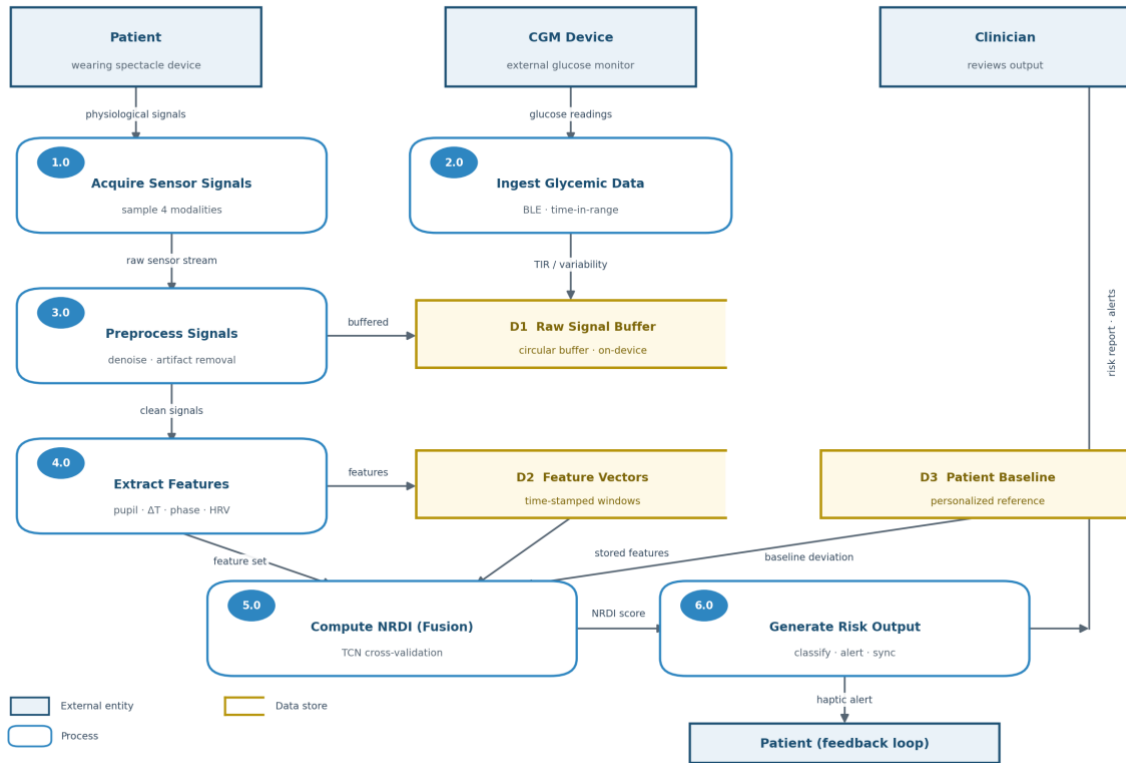


Fig. 4. Level-1 data flow diagram

Raw sensor streams enter the preprocessing stage and are buffered on device. Conditioned signals pass to feature extraction, whose outputs are both persisted and forwarded to the fusion process. The fusion process additionally consults the stored baseline to express current measurements as deviations from the individual patient reference rather than against population norms, which materially improves sensitivity to early change. The resulting index drives the output process, which classifies risk, issues local alerts to the patient and delivers reports to the clinician.

4.2.2 Use Case Diagram

Figure 5 identifies four actors interacting with the system. The patient wears and activates the device and receives haptic notifications. The health worker performs baseline calibration, initiates glucose monitor pairing and initiates index computation during screening encounters. The clinician reviews synchronised records and risk classifications. The continuous glucose monitor participates as an external system supplying glycaemic context.

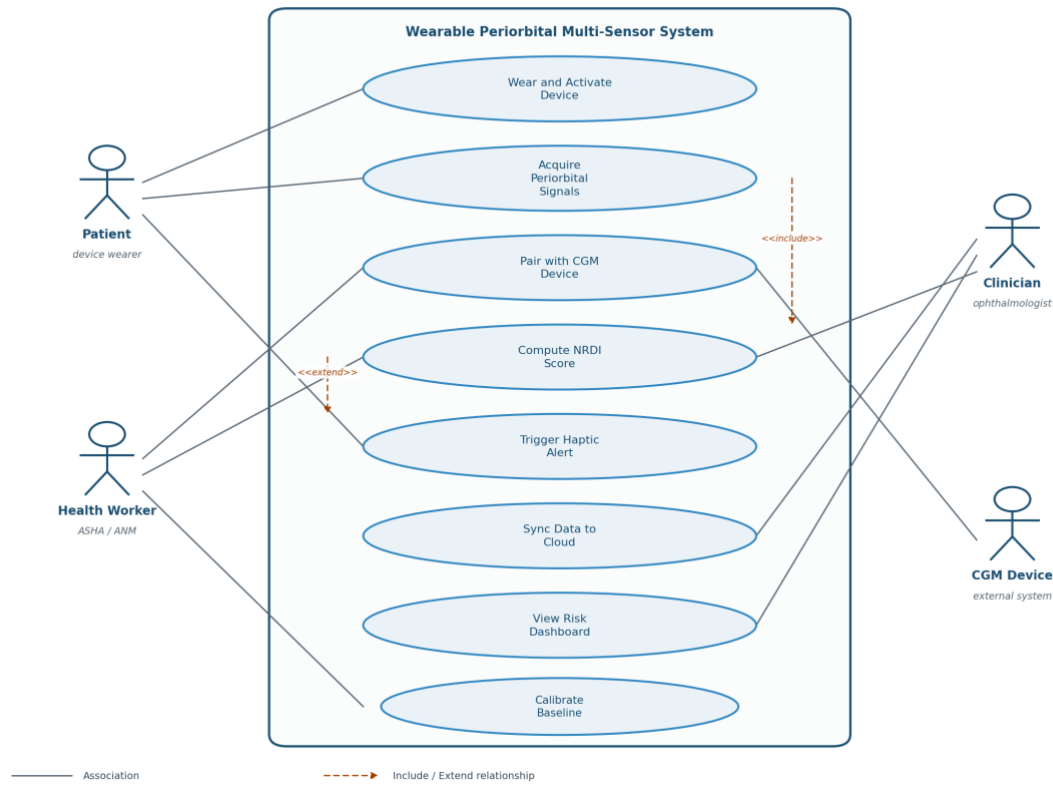


Fig. 5. Use case diagram

Signal acquisition is an included behaviour of index computation, since the index cannot be computed without a current acquisition cycle. Haptic alerting extends index computation, occurring only on the conditional branch in which the computed value exceeds the configured threshold. This structure separates the mandatory computational path from the conditional notification path, which is significant for firmware organisation and for power management, since the actuator is energised only on exception.

5. REFERENCES

Journals: (IEEE Format)

- [1] R. Chandrasekar, S. Ramesh, and K. Vasudevan, "Assessment of ocular surface temperature in diabetic retinopathy using infrared thermography," *Experimental Eye Research*, Vol. 211, (2021), pp. 108-115.
- [2] A. Sodi, S. Giambene, and U. Menchini, "Ocular surface temperature in diabetic retinopathy: A pilot study by infrared thermography," *European Journal of Ophthalmology*, Vol. 19, No. 6, (2009), pp. 1004-1008.
- [3] M. Dütsch, H. Marthol, G. Michelson, B. Neundorfer, and M. J. Hilz, "Pupillography refines the diagnosis of diabetic autonomic neuropathy," *Journal of the Neurological Sciences*, Vol. 222, No. 1-2, (2004), pp. 75-81.
- [4] G. Ferrari, M. Tedesco, and P. Rama, "Pupillary light reflex dynamics in diabetic retinal neuropathy," *Investigative Ophthalmology and Visual Science*, Vol. 61, No. 11, (2020), pp. 24-32.
- [5] S. Hwang, J. Kim, and Y. Park, "Association of bioimpedance phase angle with diabetic macular edema assessed by optical coherence tomography," *Journal of Clinical Medicine*, Vol. 12, No. 20, (2023), pp. 6511-6524.
- [6] C. Yao, L. Zhang, and H. Chen, "Fluid overload assessed by bioelectrical impedance analysis and its association with diabetic macular edema," *Diabetes Research and Clinical Practice*, Vol. 187, (2022), pp. 109-118.
- [7] ACCORD Study Group, "Cardiac autonomic dysfunction measured by heart rate variability and incident vision-threatening diabetic retinopathy," *Diabetes Care*, Vol. 45, No. 5, (2022), pp. 1156-1164.
- [8] Y. F. Ko, K. W. Kuo, and C. Y. Chen, "Design of a wearable pupillometer for the assessment of diabetic autonomic neuropathy," *Applied Optics*, Vol. 53, No. 29, (2014), pp. H27-H34.
- [9] A. Rasheed, S. Kumar, and M. Prasad, "Relationship between diabetic retinopathy and diabetic peripheral neuropathy in type 2 diabetes mellitus," *Indian Journal of Ophthalmology*, Vol. 69, No. 11, (2021), pp. 3208-3213.
- [10] S. Vujosevic, S. J. Aldington, P. Silva, C. Hernandez, P. Scanlon, T. Peto, and R. Simo, "Screening for diabetic retinopathy: New perspectives and challenges," *The Lancet Diabetes and Endocrinology*, Vol. 8, No. 4, (2020), pp. 337-347.
- [11] R. Raman, S. Sivaprasad, and T. Sharma, "Prevalence of diabetic retinopathy in India stratified by known and undiagnosed diabetes: Results from the SMART India population-based cross-sectional screening study," *The Lancet Global Health*, Vol. 10, No. 12, (2022), pp. e1764-e1773.

- [12] R. M. Anjana, M. Deepa, and R. Pradeepa, "Prevalence of diabetes and prediabetes in 15 states of India: Results from the ICMR-INDIAB population-based cross-sectional study," *The Lancet Diabetes and Endocrinology*, Vol. 5, No. 8, (2017), pp. 585-596.
- [13] V. Viswanathan and R. Mirshad, "Advancing the understanding and management of diabetic peripheral neuropathy," *International Journal of Diabetes in Developing Countries*, Vol. 44, No. 1, (2024), pp. 1-2.
- [14] R. A. Malik, M. Tavakoli, and I. N. Petropoulos, "Corneal confocal microscopy: A non-invasive surrogate of nerve fibre damage in diabetic neuropathy," *Diabetologia*, Vol. 65, No. 4, (2022), pp. 601-612.
- [15] J. Carmichael, H. Fadavi, and F. Ishibashi, "Advances in screening, early diagnosis and accurate staging of diabetic neuropathy," *Frontiers in Endocrinology*, Vol. 12, (2022), pp. 671-689.
- [16] P. H. Charlton, J. Allen, and R. Bailon, "The 2023 wearable photoplethysmography roadmap," *Physiological Measurement*, Vol. 44, No. 11, (2023), pp. 111001-111036.
- [17] TODAY Study Group, "Heart rate variability in youth-onset type 2 diabetes," *Diabetes Care*, Vol. 42, No. 10, (2019), pp. 1929-1936.
- [18] K. Soman, A. Bhaskar, and G. Devi, "Prevalence of peripheral neuropathy in type 2 diabetic adults in central Kerala: A cross-sectional study," *Indian Journal of Medical Research*, Vol. 161, No. 6, (2025), pp. 710-717.

Conference: (*IEEE Format*)

- [19] S. Chatterjee, R. Banerjee, and A. Sinha, "Multi-modal sensor fusion for non-invasive health monitoring in wearable systems," in *Proc. IEEE International Conference on Body Sensor Networks*, Boston, USA, (2023), pp. 142-148.
- [20] M. Kumar and P. Sharma, "Edge computing architectures for continuous physiological monitoring in resource-constrained settings," in *Proc. IEEE International Conference on Biomedical Engineering and Sciences*, Bengaluru, India, (2024), pp. 88-94.

Books:

- [21] J. G. Webster and A. J. Nimunkar, *Medical Instrumentation: Application and Design*, 5th ed., John Wiley and Sons, 2020.
- [22] R. B. Northrop, *Non-Invasive Instrumentation and Measurement in Medical Diagnosis*, 2nd ed., CRC Press, 2017.

Weblinks:

1. International Diabetes Federation, IDF Diabetes Atlas, <https://www.diabetesatlas.org>
2. Indian Council of Medical Research, ICMR-INDIAB Study Reports, <https://main.icmr.nic.in>

3. Analog Devices, AD5933 Impedance Converter Network Analyzer Datasheet, <https://www.analog.com>
4. Melexis, MLX90614 Infrared Thermometer Datasheet, <https://www.melexis.com>